

Multi-Dot Matrix Scrolling Message Display Using 8051

Project Description:

Design and simulate a scrolling LED message board using the 8051 microcontroller and multiple 8×8 dot matrix displays connected together. The system should display long scrolling messages continuously across the displays similar to commercial LED advertising boards.

Project Requirements:

1. Use at least:
 - 4 dot matrix displays (minimum)
2. The message should scroll smoothly from right to left across all displays.
3. Implement a complete character lookup table for:
 - Uppercase and Lowercase letters
 - Digits 0–9
 - Space character
4. Use timer interrupts for display refreshing and scrolling.
5. The display must support long strings, for example:
“WELCOME TO ANTALYA BILIM UNIVERSITY SPRING 2026”
6. Use multiplexing correctly to drive all displays.
7. Add at least TWO control features using push buttons or DIP switches:
 - Increase scrolling speed
 - Decrease scrolling speed
 - Pause/Resume
 - Change message
 - Reverse scrolling direction
8. The project must be implemented in:
 - Proteus simulation
 - 8051 C or Assembly language
9. Students must write modular code:
 - Display driver module
 - Character lookup table module
 - Scrolling control module

Recommended Hardware/Components:

- AT89C51 or compatible 8051

- Multiple 8×8 LED dot matrices
- 74HC595 shift registers (recommended)
- Push buttons or DIP switches

Advanced Bonus Features:

- Bidirectional scrolling
- Animated text effects

Expected Learning Outcomes:

Students will practice:

- Multiplexed display systems
- Timer interrupts
- Embedded memory management
- Lookup table design
- Large-scale I/O interfacing
- Real-time embedded programming